SIEM01: Rebuilding the SIEM After a Hardware Failure

Purpose: Documents the recovery of the lab's log collection capability after the original Wazuh host failed, the hardware role swap that made the rebuild better than the original, and the Wazuh deployment on the new host. Written so a reviewer can see the reasoning behind the placement decision, not only the installation steps.

Host: `SIEM01`, Dell OptiPlex 9020, Ubuntu 24.04.2 LTS, `192.168.20.2/24` on VLAN 20 (BlueTeam), switch Port 4.

Status: Host live and Wazuh 4.14.7 installed 2026-09-05. No agents deployed yet.

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1. What failed, and the decision that followed

The original SIEM was a physical machine running Rocky Linux 9.6 at `192.168.20.2`. It lost power on 2026-08-29 and did not return. That was the second hardware failure in this lab, after a server NIC, and it left the environment with no central log collection: pfSense continued logging locally, but nothing was being shipped or correlated.

The obvious response was to rebuild the same thing on the same kind of hardware. A better question was which machine *should* hold this role.

At that point the estate had two physical Linux hosts and a hypervisor with spare capacity. The monitoring host was a Dell OptiPlex 9020 with 8 cores and 16 GB of RAM, running Grafana and Prometheus, a workload that comfortably fits in about 2 GB. Meanwhile the service that needed rebuilding, Wazuh, ships an Indexer (an OpenSearch instance) whose practical memory floor is around 8 GB, and which accumulates data that must survive.

So the roles were the wrong way round, and the failure was the opportunity to correct it:

	Before	After
Dell OptiPlex 9020 (physical)	Grafana and Prometheus, VLAN 60	**`SIEM01`**, Wazuh, VLAN 20
Proxmox guest	none	`MON01`, Grafana and Prometheus, VLAN 60

The principle is worth stating plainly, because it generalises: **put the memory-hungry, data-bearing service on dedicated hardware, and the light, reproducible one on the hypervisor.** Grafana and Prometheus can be rebuilt from a configuration file in minutes. A SIEM's value is its accumulated history and the fact that it keeps running while everything around it is being investigated.

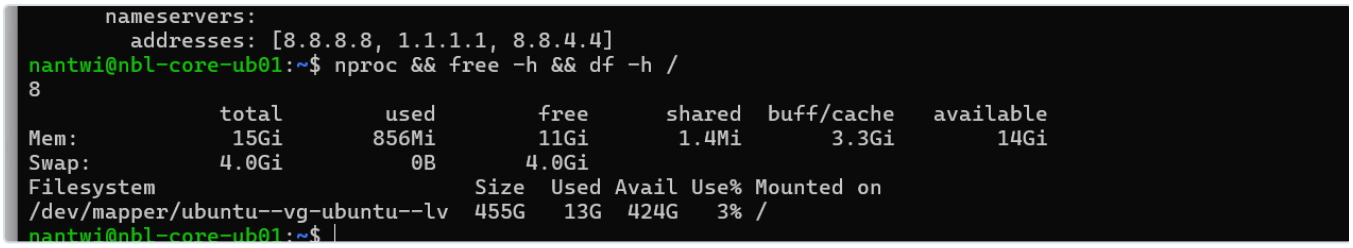

Figure 16.1: The candidate host: 8 cores, 15 GiB usable RAM, and 424 GB free of a 455 GB volume. The disk figure matters as much as the memory, because the Indexer is what eventually fills a SIEM's storage.

Full rationale and the naming decisions are in `docs/14` section 5.3.

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2. Repurposing the host in place

The machine was **not** reinstalled. Ubuntu 24.04 carried over, which means Wazuh moved from Rocky Linux to Ubuntu. Three changes moved the host from one role to the other.

Renamed to match the convention (`docs/14` section 5.1):

```
sudo hostnamectl set-hostname SIEM01
sudo sed -i "s/nbl-core-ub01/SIEM01/g" /etc/hosts
```

The old name, `nbl-core-ub01` , encoded the operating system and nothing about the machine's purpose. That is precisely the failure mode the naming convention exists to prevent.

Readdressed from VLAN 60 to VLAN 20, and **recabled** from switch Port 8 to Port 4, the untagged VLAN 20 access port.

The cloud-init trap

The netplan file on an Ubuntu Server install is named `50-cloud-init.yaml` , and that name is a warning. It is generated by **cloud-init**, the program that configures a fresh machine on first boot. A file written by a program can be rewritten by that program, and a manual edit would be silently reverted at the next boot, leaving the host on its old address on a port that no longer carries that VLAN.

Disabling cloud-init's network management first is what makes the edit durable:

```
echo 'network: {config: disabled}' | sudo tee /etc/cloud/cloud.cfg.d/99-disable-network-config.cfg
```

The configuration then uses the current netplan syntax. `gateway4:` is deprecated in 24.04 and replaced by a `routes:` entry, and DNS points at pfSense rather than a public resolver, matching DC01 and KALI01:

```
network:
  version: 2
  ethernets:
    eno1:
```

```
      dhcp4: no
      addresses:
        - 192.168.20.2/24
      routes:
        - to: default
          via: 192.168.20.1
      nameservers:
        addresses: [192.168.20.1]
```

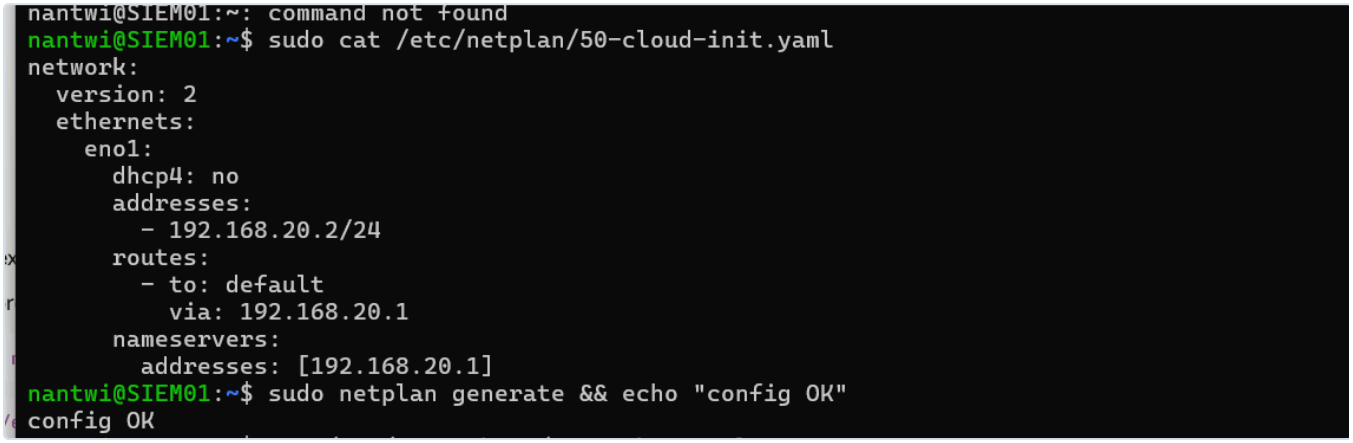

Figure 16.2: The VLAN 20 configuration after a reboot, with `netplan generate` confirming it parses. That it survived the reboot is the evidence that cloud-init is genuinely no longer managing the interface.

The general lesson: on Linux, a configuration file named after a program is usually owned by that program. Editing it works until the program runs again. Either turn the program off, as here, or place your settings in a higher-numbered file that wins.

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3. Two failures worth recording

Applying the network change before moving the cable. `netplan apply` was run while the machine was still plugged into the VLAN 60 port. The host immediately held an address that did not exist on the network it was attached to: unreachable, with no route to its gateway. Nothing was damaged, and moving the cable resolved it, but the correct order is to write the configuration, shut down, move the cable, then power on. A machine should never be left holding an address its physical connection cannot serve.

SSH refused the connection afterwards. The address `192.168.20.2` had belonged to the failed Rocky Linux host, whose SSH host key was still recorded in the administrator's `known_hosts`. A different physical machine now answered at that address with a different key, so SSH refused:

```
WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!
```

This is the control working exactly as designed. SSH cannot distinguish "the hardware was replaced" from "something is intercepting this connection", so it stops and requires a human decision. The response was to verify rather than dismiss: the fingerprint was read directly from the host itself,

```
sudo ssh-keygen -lf /etc/ssh/ssh_host_ed25519_key.pub
```

confirmed identical to the one SSH had presented, and only then was the stale record removed with `ssh-keygen -R 192.168.20.2`.

SSH also volunteered corroboration, noting the same key was already known under `192.168.60.2`, the machine's previous address. That is independent confirmation that this was a host that moved, not a stranger.

The distinction that matters: deleting the warning and reconnecting is what most people do, and it works. Verifying the fingerprint out of band is what turns a dismissed alert into a cleared one. If that warning ever appears when nothing has changed, it is the one time it is telling you something real.

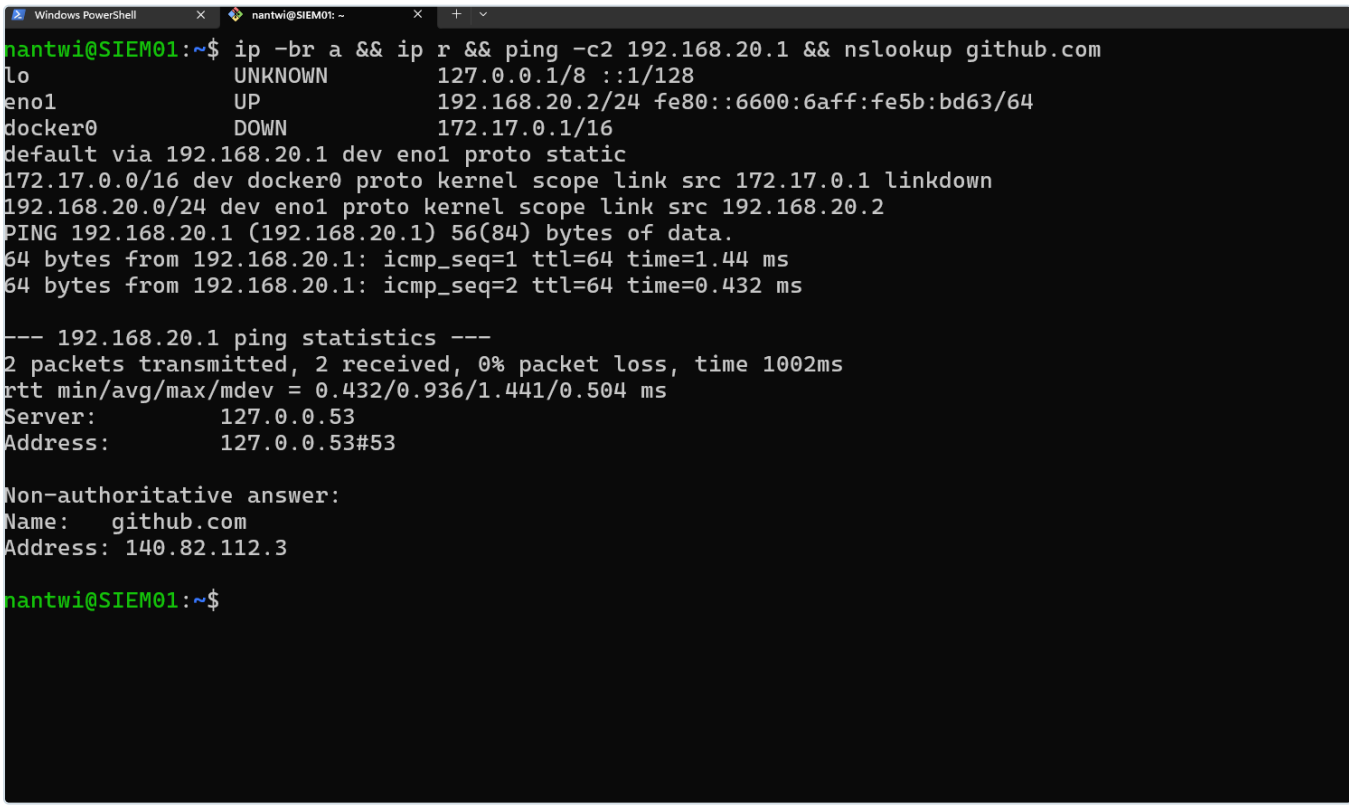

Figure 16.3: Post-reboot verification: the static address held, the default route is marked `proto static`, the gateway answers, and name resolution works. Four separate things confirmed in one command.

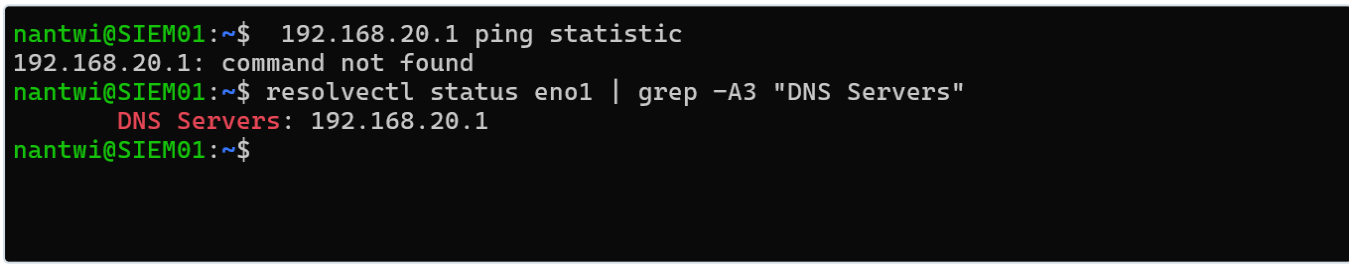

Figure 16.4: `resolvectl` confirming the upstream resolver is `192.168.20.1`, pfSense, rather than a leftover public server. `127.0.0.53` in an `nslookup` result is systemd-resolved, a local stub that forwards on; it is worth checking what sits behind it.

4. Firewall consequences

Two rules were required before the host could be administered, and both were built before the move rather than after.

`MGMT-11` permits the management plane to administer the SIEM, using aliases so the rule reads by intent rather than by number:

Alias	Type	Value
`SIEM01_HOST`	Host	`192.168.20.2`
`SIEM_ADMIN`	Ports	`22` (SSH), `443` (Wazuh dashboard)

Port 443 was included before Wazuh existed, on the expectation that the dashboard would use it. The installer confirmed this during the run (`Wazuh web interface port will be 443`), so no rule change was needed afterwards.

The rule is **logged**, on the same reasoning as `MGMT-08` on the domain controller: administrative access to a machine holding security evidence is privileged activity and has to be attributable (NIST AC-17, AU-2). Registered in `docs/13`.

The DNS Resolver check. Before moving the host, the pfSense DNS Resolver was confirmed to be listening on the BLUETEAM interface. This is the third time in this build that a service bound to the wrong interfaces would have produced a correct-looking configuration and a silent failure, after the same issue on VLAN 30 (`docs/15` section 6). It is now a standing pre-flight check when a host moves to a new segment.

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5. Installing Wazuh

Wazuh is three programs, and knowing which is which is the difference between diagnosing a problem and guessing at one:

Component	Job	Listens on
Wazuh Indexer	Stores and searches events. OpenSearch underneath. The memory-hungry component.	9200 (localhost)
Wazuh Server	The manager. Agents connect to it; it applies detection rules and forwards results.	1514, 1515, 55000
Wazuh Dashboard	The web interface.	443
(Filebeat)	Ships the manager's own alerts into the Indexer.	n/a

"Wazuh is down" is never a useful diagnosis. It is one of these.

The all-in-one installer deploys all three on a single host, which suits 16 GB and 8 cores. A distributed deployment only earns its complexity when the Indexer outgrows one machine.

The agent and manager conflict

The first install attempt was refused:

```
ERROR: Wazuh manager already installed.
```

Investigation showed no manager, only a `wazuh-agent` left from when this machine was the monitoring host reporting to the old SIEM.

A manager and an agent cannot coexist on one host, because both packages own `/var/ossec`. It is not a limitation to work around: the manager monitors its own host natively, so an agent on the manager would be redundant as well as conflicting.

```
nantwi@SIEM01:~$  192.168.20.1 ping statistic
192.168.20.1: command not found
nantwi@SIEM01:~$ resolvectl status eno1 | grep -A3 "DNS Servers"
       DNS Servers: 192.168.20.1
nantwi@SIEM01:~$ curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-install.sh -a
[sudo] password for nantwi:
05/09/2026 19:51:26 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.7
05/09/2026 19:51:26 INFO: Verbose logging redirected to /var/log/wazuh-install.log
05/09/2026 19:51:27 ERROR: Wazuh manager already installed.
05/09/2026 19:51:27 INFO: If you want to overwrite the current installation, run this script adding the option -o/--overwrite
. This will erase all the existing configuration and data.
nantwi@SIEM01:~$ ls
 bluelab-ansible   Docker-Lessons   test_winrm.py  'udo systemctl restart grafana-server'   wazuh-install.sh
nantwi@SIEM01:~$ sudo bash ./wazuh-install.sh -a
05/09/2026 19:52:02 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.7
05/09/2026 19:52:02 INFO: Verbose logging redirected to /var/log/wazuh-install.log
05/09/2026 19:52:03 ERROR: Wazuh manager already installed.
05/09/2026 19:52:03 INFO: If you want to overwrite the current installation, run this script adding the option -o/--overwrite
. This will erase all the existing configuration and data.
nantwi@SIEM01:~$
```

Figure 16.5: The installer refusing, and the investigation that followed. Only `wazuh-agent 4.11.2` was present, a leftover from the machine's previous role.

The installer offers `-o/--overwrite`, which erases all existing Wazuh configuration and data. Removing the single package explicitly was preferred, because the overwrite flag is a blunt instrument and on a machine whose job is to be trustworthy it is worth knowing exactly what changed:

```
sudo systemctl disable --now wazuh-agent
sudo apt purge -y wazuh-agent
sudo rm -rf /var/ossec
```

```
nantwi@SIEM01:~$ sudo systemctl disable --now wazuh-agent && sudo apt purge -y wazuh-agent && sudo rm -rf /var/ossec
Removed "/etc/systemd/system/multi-user.target.wants/wazuh-agent.service".
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages will be REMOVED:
  wazuh-agent*
0 upgraded, 0 newly installed, 1 to remove and 0 not upgraded.
After this operation, 39.7 MB disk space will be freed.
(Reading database ... 133547 files and directories currently installed.)
Removing wazuh-agent (4.11.2-1) ...
(Reading database ... 133141 files and directories currently installed.)
Purging configuration files for wazuh-agent (4.11.2-1) ...
nantwi@SIEM01:~$ dpkg -l | grep -i wazuh; ls /var/ossec 2>&1; echo "--- clean if both empty ---"
ls: cannot access '/var/ossec': No such file or directory
--- clean if both empty ---
nantwi@SIEM01:~$ free -h && sudo ss -tlnp | grep -E ':(443|1514|1515|9200|55000)\s'; echo "--- ports clear if nothing above -
--"
               total        used        free      shared  buff/cache   available
Mem:            15Gi       624Mi        14Gi       1.4Mi       930Mi        14Gi
Swap:          4.0Gi          0B       4.0Gi
--- ports clear if nothing above ---
nantwi@SIEM01:~$ sudo bash ./wazuh-install.sh -a
05/09/2026 19:58:03 INFO: Starting Wazuh installation assistant. Wazuh version: 4.14.7
05/09/2026 19:58:03 INFO: Verbose logging redirected to /var/log/wazuh-install.log
05/09/2026 19:58:12 INFO: Verifying that your system meets the recommended minimum hardware requirements.
05/09/2026 19:58:12 INFO: Wazuh web interface port will be 443.
05/09/2026 19:58:16 INFO: --- Dependencies ----
05/09/2026 19:58:16 INFO: Installing debhelper.
```

Figure 16.6: The agent purged, `/var/ossec` gone, Wazuh's ports confirmed free, and the installer proceeding past the check that had blocked it.

Running the install

The installer was downloaded to a file rather than piped into a shell:

```
curl -sO https://packages.wazuh.com/4.14/wazuh-install.sh
sudo bash ./wazuh-install.sh -a
```

The pattern `curl ... | sudo bash` is common and is worth avoiding on principle. It executes code as root that was never seen, from a server that could have been compromised between publication and retrieval. Downloading first costs nothing and makes inspection possible.

```
Windows PowerShell   nantwi@SIEM01: ~
05/09/2026 19:59:46 INFO: wazuh-indexer service started.
05/09/2026 19:59:46 INFO: Initializing Wazuh indexer cluster security settings.
05/09/2026 19:59:52 INFO: Wazuh indexer cluster security configuration initialized.
05/09/2026 19:59:52 INFO: Wazuh indexer cluster initialized.
05/09/2026 19:59:52 INFO: --- Wazuh server ---
05/09/2026 19:59:52 INFO: Starting the Wazuh manager installation.
05/09/2026 20:01:08 INFO: Wazuh manager installation finished.
05/09/2026 20:01:09 INFO: Wazuh manager vulnerability detection configuration finished.
05/09/2026 20:01:09 INFO: Starting service wazuh-manager.
05/09/2026 20:01:25 INFO: wazuh-manager service started.
05/09/2026 20:01:25 INFO: Starting Filebeat installation.
05/09/2026 20:01:48 INFO: Filebeat installation finished.
05/09/2026 20:01:52 INFO: Filebeat post-install configuration finished.
05/09/2026 20:01:52 INFO: Starting service filebeat.
05/09/2026 20:01:53 INFO: filebeat service started.
05/09/2026 20:01:53 INFO: --- Wazuh dashboard ---
05/09/2026 20:01:53 INFO: Starting Wazuh dashboard installation.
05/09/2026 20:10:38 INFO: Wazuh dashboard installation finished.
05/09/2026 20:10:38 INFO: Wazuh dashboard post-install configuration finished.
05/09/2026 20:10:38 INFO: Starting service wazuh-dashboard.
05/09/2026 20:10:38 INFO: wazuh-dashboard service started.
05/09/2026 20:10:40 INFO: Updating the internal users.
05/09/2026 20:10:45 INFO: A backup of the internal users has been saved in the /etc/wazuh-indexer/internalusers-backup folder
.
05/09/2026 20:10:54 INFO: The filebeat.yml file has been updated to use the Filebeat Keystore username and password.
05/09/2026 20:11:25 INFO: Initializing Wazuh dashboard web application.
05/09/2026 20:11:27 INFO: Wazuh dashboard web application initialized.
05/09/2026 20:11:27 INFO: --- Summary ---
05/09/2026 20:11:27 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
```

Figure 16.7: The Indexer, manager, Filebeat and Dashboard each installing and starting in sequence. This screenshot is deliberately cropped: the installer prints the generated admin credentials immediately below, and they must not reach a repository.

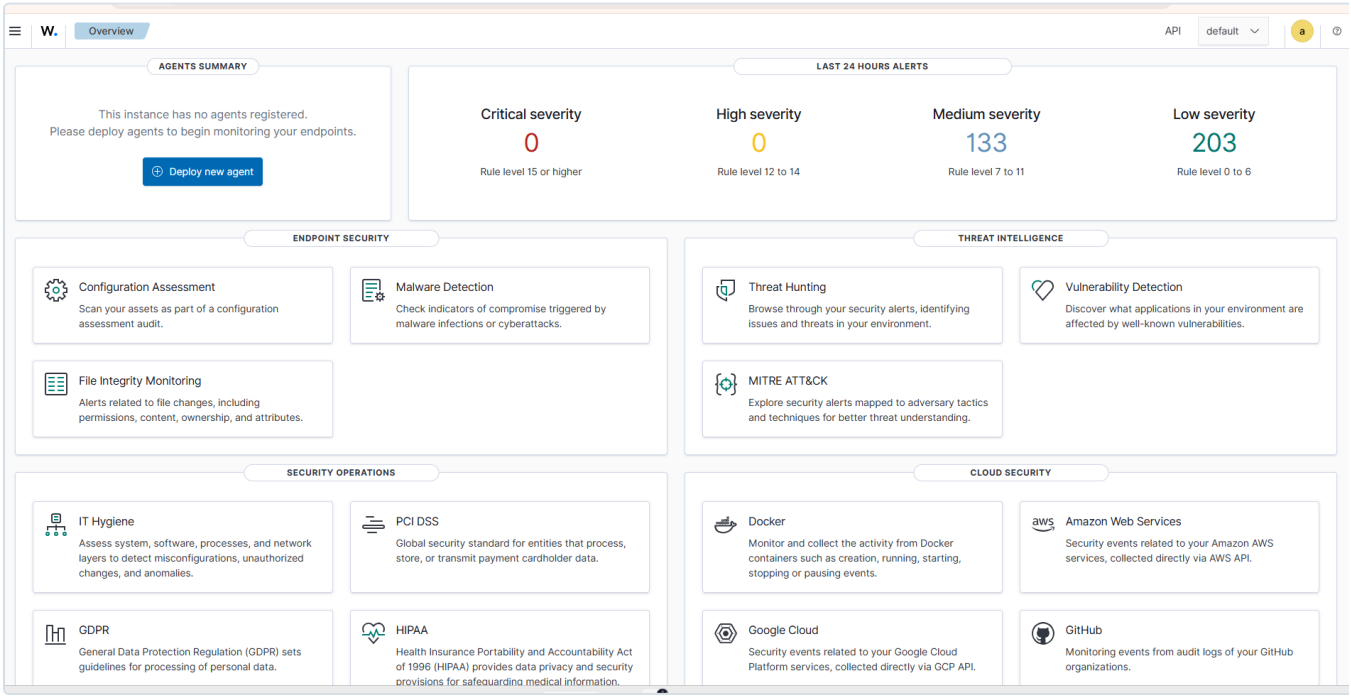

Figure 16.8: The dashboard on first login. "No agents registered" alongside 336 alerts is not a contradiction: the manager monitors its own host as agent `000`.

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6. Credentials, and why the dashboard could not change them

The installer prints a generated `admin` password once. Attempting to change it through the dashboard fails:

```
Failed to reset password. {"status":"FORBIDDEN","message":"Resource 'admin' is reserved."}
```

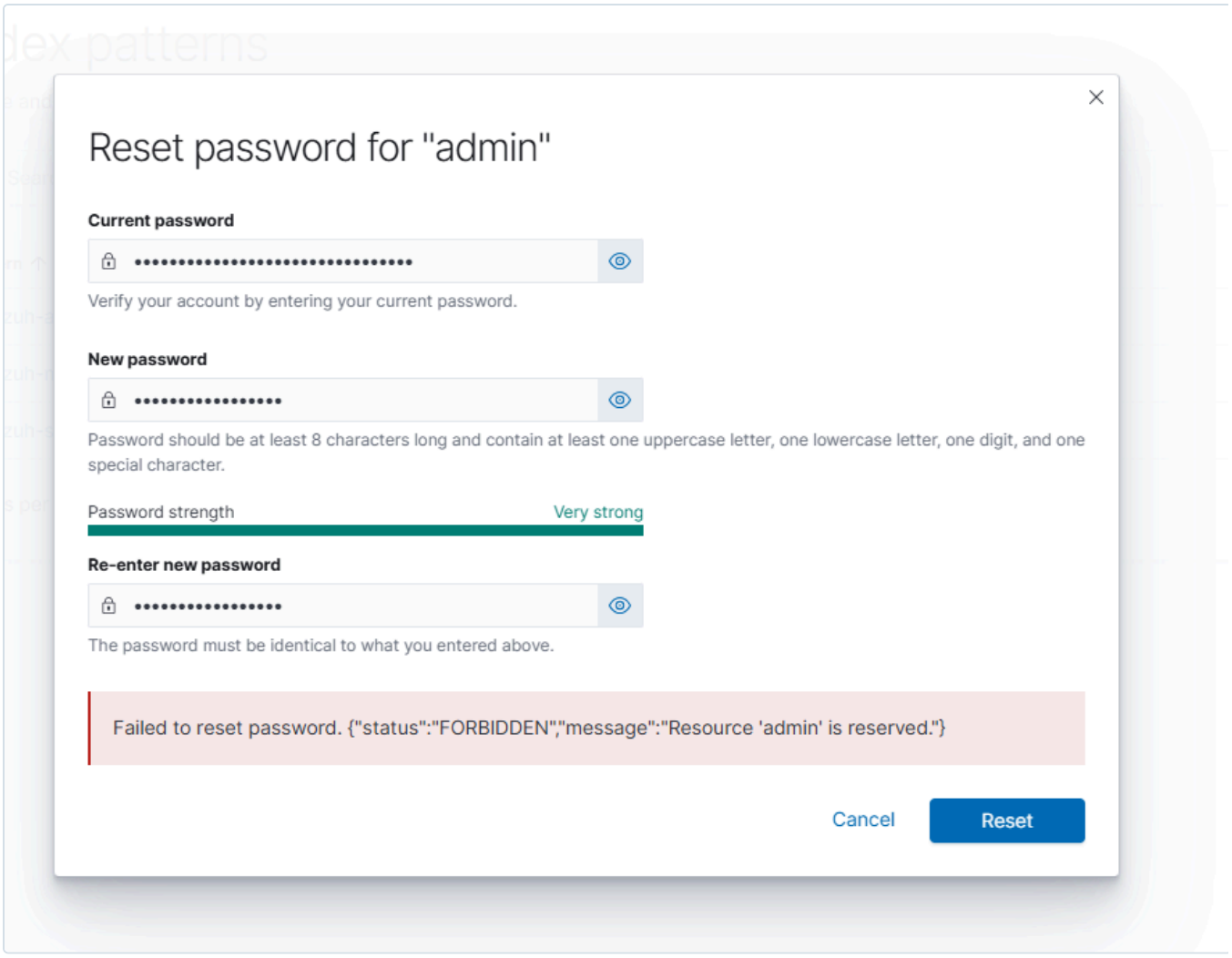

Figure 16.9: The dashboard refusing to change its own administrative password. The account is a reserved resource in the Indexer's security configuration and cannot be modified through the API the interface uses.

This is a deliberate boundary rather than a defect. The `admin` account is defined in the security plugin's `internal_users.yml` and marked **reserved**, which makes it immutable from the application layer. A compromised dashboard session therefore cannot lock the owner out of their own cluster. Changing it requires filesystem access to the host, which is a meaningfully higher bar:

```
curl -sO https://packages.wazuh.com/4.14/wazuh-passwords-tool.sh
sudo bash wazuh-passwords-tool.sh -u admin -p '<new password>'
```

The tool rewrites the hash and reruns `securityadmin` to push the change into the cluster.

An evidence-handling rule came out of this. The moment an installer prints a generated credential is exactly the moment people screenshot it. Figure 16.7 is cropped for that reason. Every screenshot is now checked for passwords, keys and tokens before it enters `images/`, and a credential that has appeared in a screenshot is rotated rather than trusted.

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7. What it monitors today

With no agents deployed, the dashboard already shows several hundred alerts. They all carry agent ID `000`, which is always the manager itself.

Most are **Security Configuration Assessment** findings: Wazuh ships CIS Benchmark policies and evaluates the host against them at startup. These are not detections of an attack. They are an audit of the machine's own hardening, and they are useful for two reasons.

The first is direct: a SIEM holds evidence of everything else in the environment, which makes it a high-value target, and the tool has just produced a prioritised list of its own weaknesses.

The second is conceptual. SCA is continuous configuration monitoring, which is the same control family being satisfied by hand for pfSense in `docs/13`: NIST CM-2 (Baseline Configuration) and CM-6 (Configuration Settings). The firewall register is a manual implementation of what SCA does automatically for hosts.

Planned as an exercise: record the current SCA score, harden a set of failing checks, and show the score improving. That produces a before-and-after with evidence, in the same shape as the management-plane isolation test in `docs/13` section 3.1.

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8. What comes next, and where the rules go

Deploying the first agent is where the firewall model becomes concrete, and it is the point most people get backwards.

A pfSense rule governs who may start a conversation. Replies return through the state table and never need a rule. So the question for any service is not "who should reach this server?" but "who initiates?"

Wazuh agents connect **outbound** to the manager. The manager listens. Therefore:

To achieve this	The rule belongs on	Because
DC01 reports to Wazuh	**ENTERPRISELAN (VLAN 50)**	DC01 initiates, and it lives on VLAN 50
KALI01 reports to Wazuh	**REDTEAM (VLAN 30)**	KALI01 initiates
Administrator opens the dashboard	**MANAGEMENT (VLAN 10)**	the workstation initiates (`MGMT-11`, built)

Almost nothing is added to the BLUETEAM tab itself. The instinct is to grant the SIEM broad reach on the reasoning that it must "see everything". That is backwards twice over: it does not grant what agents actually need, and it would give the machine holding all the security evidence a standing path across the entire estate.

The agent rules are `ENT-05` and `RED-04`, permitting TCP 1514 (events) and 1515 (enrollment) to `SIEM01_HOST`. They are registered in `docs/13` as they are built.

Also outstanding: Grafana and Prometheus are still installed on this host and should be removed once `MON01` exists, and Docker was removed as unused attack surface on a machine that has no need of it.

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9. Standards alignment

Practice in this document	Standard
Centralised collection and retention of security events	**NIST SP 800-53 AU-6** (Audit Record Review), **AU-9** (Protection of Audit Information)
Continuous configuration assessment against a benchmark	**NIST SP 800-53 CM-2, CM-6**; **CIS Controls v8, Control 4**
Administrative access to the SIEM restricted and logged	**NIST SP 800-53 AC-17, AU-2**
Recovery of a failed service onto sound hardware, with placement justified	**NIST SP 800-53 CP-10** (System Recovery)
Unnecessary software removed from a security-critical host	**NIST SP 800-53 CM-7** (Least Functionality)
Host identity verified out of band before trust was re-established	**NIST SP 800-53 IA-3** (Device Identification and Authentication)
Role-based naming supporting asset inventory	**NIST SP 800-53 CM-8**
Segmentation between the monitored estate and the monitoring segment	**NIST SP 800-53 SC-7**, **CIS Controls v8, Control 12**

Related documents

- `docs/13-firewall-rulebase-governance.md`, the rule register including `MGMT-11` and the pending agent rules
- `docs/14-proxmox-storage-backup-capacity.md`, the naming convention and the role swap rationale
- `docs/15-kali-attack-host-build.md`, the same interface-direction reasoning applied to the attack segment
- `docs/02-security-monitoring.md`, the original Wazuh deployment on the failed Rocky Linux host